

Supplementary Material for “Is Prompt Rewriting Always Beneficial? A Benchmark for Quantifying Prompt Informativeness and Stability in Text-to-Image Models”

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Additional Details on Stability Modeling and AP Bench

Scope of the Information-Theoretic Interpretation

The information-theoretic interpretation below motivates the stability metrics used in our evaluation. Because the true intent posterior is inaccessible, the resulting quantities should be understood as computable surrogates rather than exact mutual-information estimates.

CLIP Target-Category Probability

Information-Theoretic Motivation. Let $Y \in \mathcal{Y}$ denote the discrete user intent category and $X \in \mathcal{X}$ the generated image. The mutual information between user intent and generated images is defined as

$$I(Y; X) = H(Y) - H(Y | X),$$

where $H(\cdot)$ denotes Shannon entropy. Since the prior distribution over intent categories is fixed across prompts and models, $H(Y)$ can be treated as a constant. Therefore, maximizing $I(Y; X)$ is equivalent to minimizing the conditional entropy $H(Y | X)$. Although the true posterior $p(Y | X)$ is intractable, categorical recoverability provides a useful theoretical motivation. Let $q_{\text{CLIP}}(c | X)$ denote the normalized CLIP confidence for category c , and define the corresponding hard prediction as

$$\hat{Y} = \arg \max_{c \in \mathcal{Y}} q_{\text{CLIP}}(c | X).$$

The classification error probability is

$$P_e = \Pr(\hat{Y} \neq Y).$$

By Fano’s inequality, the conditional entropy is upper-bounded as

$$H(Y | X) \leq h(P_e) + P_e \log(|\mathcal{Y}| - 1), \quad (1)$$

where $h(\cdot)$ denotes the binary entropy function. This relation motivates using target-category recoverability as an operational signal of intent preservation: lower hard classification error yields a tighter information-theoretic guarantee. We use this argument only as motivation and do not use hard classification accuracy as the reported metric.

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Reported Soft Probability and Generation-to-Generation Variation. Following the entity–template expansion protocol used in the main paper, we generate n images $\{X_1, \dots, X_n\}$ for each prompt. For image X_j , let

$$a_j = q_{\text{CLIP}}(Y_c | X_j) \in [0, 1]$$

denote the normalized CLIP probability assigned to the target intent category Y_c . We report the empirical mean and standard deviation

$$\begin{aligned} \text{Acc}_{\text{mean}}(P) &= \frac{1}{n} \sum_{j=1}^n a_j, \\ \text{Acc}_{\text{std}}(P) &= \left[\frac{1}{n} \sum_{j=1}^n (a_j - \text{Acc}_{\text{mean}}(P))^2 \right]^{1/2}. \end{aligned}$$

The mean measures expected target-intent recoverability, whereas the standard deviation measures how much this soft confidence varies across stochastic generations. Thus, unlike the variance of a Bernoulli correctness indicator, the reported standard deviation is not determined by the mean and provides a distinct measure of generation-to-generation dispersion.

We do not claim that the reported CLIP score is an unbiased estimator of $I(Y; X)$; it is an operational proxy for intent recoverability motivated by the information-theoretic formulation.

MLLM-Based Semantic Alignment

To capture fine-grained semantic correspondence, we use a multi-modal large language model (MLLM) to score each generated image X_j against a short textual description of the user intent Y , producing $s_j \in [0, 1]$. For a prompt P , the reported score is

$$S(P) = \frac{1}{n} \sum_{j=1}^n s_j.$$

It evaluates whether the generated images preserve the objects, attributes, relations, and other intent-bearing details in the user intent description.

The MLLM score is used as an operational semantic-alignment proxy rather than a calibrated likelihood or a direct estimator of mutual information. It complements the

category-level CLIP measure by evaluating fine-grained intent attributes that cannot be represented by a fixed class taxonomy. Accordingly, differences in this score are interpreted as differences in model-assessed semantic alignment, without assigning the score a probabilistic or variational-bound interpretation.

Prompt Entropy and Prompt Energy

Consider the Markov chain $Y \rightarrow P \rightarrow X$, where Y is user intent, P is the prompt, and X is the generated image. By the data-processing inequality:

$$I(Y; X) \leq I(Y; P),$$

indicating that the maximum achievable stability is constrained by the prompt information content.

Operationally, for a prompt $P = (w_1, \dots, w_T)$, we compute the token-average surprisal under an external pretrained language model:

$$S_\theta(P) = -\frac{1}{T} \sum_{t=1}^T \log p_\theta(w_t \mid w_1, \dots, w_{t-1}).$$

We refer to this quantity as the prompt-entropy score for brevity, but it is more precisely a model-relative token-average surprisal, or cross-entropy proxy, rather than the intrinsic entropy of an individual prompt. It measures how predictable the observed token sequence is under the scoring model and therefore provides a diagnostic of linguistic familiarity or uncertainty.

A low value does not necessarily imply that a prompt is semantically complete or faithful to the user intent: a common but underspecified prompt may also receive low surprisal. Conversely, a detailed prompt containing rare but relevant concepts may receive a high value. We therefore treat prompt entropy as a model-relative auxiliary diagnostic rather than an absolute prompt-quality metric, and interpret it jointly with prompt energy and image-side alignment measures.

Derivation of Prompt Energy. In classical statistical mechanics, a system state $x_t^{(i)}$ follows a Boltzmann distribution:

$$p(x_t^{(i)}) = \frac{\exp(-E_t^{(i)}/k\tau)}{Z_t}.$$

An autoregressive LM with parameters θ defines the probability of token x_t as

$$p_\theta(x_t \mid x_{<t}) = \frac{\exp(z_t(x_t))}{\sum_{v \in \mathcal{V}} \exp(z_t(v))},$$

where $z_t(v)$ is the logit of token v .

Logits can be related to negative energies only up to a context-dependent additive constant C_t :

$$z_t(v) = -\frac{1}{k\tau} E_t(v) + C_t.$$

Using the conventional scale $k\tau = 1$, we define the selected-token energy score

$$\tilde{e}_t(x_t) = -z_t(x_t),$$

which equals the corresponding energy only up to the context-dependent additive constant. The sequence-level score is

$$\tilde{E}_\theta(P) = -\frac{1}{T} \sum_{t=1}^T z_t(w_t).$$

This quantity is an operational proxy for the scoring model’s familiarity with, or confidence in, the selected tokens. It is not a calibrated physical energy, a strict measure of epistemic uncertainty, or an absolute measure of prompt quality.

Comparison Scope. Because logits are defined only up to additive offsets and their scales differ across language models, absolute energy values must not be compared across Llama-, Mistral-, and DeepSeek-based scorers. We compare Original, Promptist, PAG, and NoxEye only within each language-model column, using the same scoring model and pipeline. Prompt energy is interpreted jointly with the surprisal and image-side alignment diagnostics rather than as a stand-alone quality measure.

Validation on Matched Ambiguous–Clean Prompt Pairs

We validate the prompt-side entropy and energy diagnostics on more than 42,000 matched ambiguous–clean prompt pairs from ATM (Du et al. 2023). Because each pair represents a controlled contrast between an ambiguous prompt and a corresponding cleaner formulation, this evaluation tests whether the diagnostics respond in the expected direction without treating either score as an absolute measure of semantic quality. Clean prompts exhibit lower entropy and lower energy in more than 92% of the matched comparisons. This result supports their use as complementary indicators of linguistic ambiguity and model compatibility, while the exceptions reinforce the need to interpret both scores jointly with image-side intent-preservation measures.

AP Bench Data Collection

Inspired by ImageNet (Russakovsky et al. 2015), we use its 1,000 semantic categories as the initial intent taxonomy and collect authentic intent–prompt–image triplets from Lexica (<https://lexica.art/>) and Civitai (<https://civitai.com/>). Each retained triplet consists of an observable intent signal associated with a real user prompt, the prompt itself, and the image publicly paired with that prompt on the source platform. These are existing records shared on the two text-to-image communities. We do not interpret public posting as a direct or independently verified statement of user satisfaction.

For each source record, we retain the original URL and the available generation parameters and auxiliary metadata, including the seed, sampler, model, and negative prompt when provided by the platform. ImageNet serves only as the initial taxonomy for organizing semantic coverage. We additionally collect authentic samples whose intents fall outside the 1,000 predefined categories, so AP Bench is not restricted to closed-set object recognition.

Annotation and Verification Protocol

Intent in AP Bench is operationalized from the observable signals associated with each authentic platform record, principally its prompt and available context; it is not treated as a complete account of the user’s latent intent. Nor is it assigned solely from an ImageNet class or retrospectively inferred from the image by an automated evaluator. We distinguish the coarse *class-level intent* used to organize the initial taxonomy from the more detailed *natural-language user intent* expressed through the prompt and its associated record. The latter may specify objects, attributes, relations, composition, and style beyond a single category label.

Following manual review and re-verification, we stratify prompts into *novice* and *expert* subsets according to their apparent prompt-design proficiency. The categorization considers prompt specificity, completeness, structural complexity, and the use of generation-related syntax. Reviewers further verify the correspondence among the recorded intent, prompt, and image, and filter the collected content for ethical compliance, safety, and the absence of sensitive material. CLIP and MLLM scores are used only as evaluation signals and do not replace this review.

Benchmark Composition and Statistics

The resulting collection contains 17,580 authentic intent–prompt–image triplets from Lexica and Civitai. Each sample links an observable user-intent signal, the corresponding prompt, and its publicly associated image, together with the available source URL, generation parameters, and auxiliary metadata. The image–prompt association is treated as evidence that the pair was selected for sharing, rather than as a verified satisfaction label.

From the full collection, 2,048 representative samples are selected for benchmark evaluation, including 256 samples whose intents fall outside the ImageNet-1K taxonomy. AP Bench defines three difficulty levels: *Easy*, where the intent is explicit and its category belongs to the predefined taxonomy; *Medium*, where the intent is explicit but falls outside that taxonomy; and *Hard*, where the category belongs to the taxonomy but the user intent is expressed ambiguously. The benchmark therefore combines broad class-anchored coverage with open-set intents while preserving the distinction between coarse retrieval categories and detailed user intent.

Implementation Details and Additional Experiments

Training-Data Construction

NoxEye is trained in two stages. For the information enhancer, we select high-quality prompt–image pairs from the collected data while excluding every AP Bench evaluation sample. Gemini 2.5 (Team et al. 2025) and GPT-4o (OpenAI et al. 2024) generate multiple candidate refinements for each user prompt. Human experts evaluate the candidates for (i) semantic faithfulness to the original prompt and intended user intent, (ii) absence of hallucinated or unsupported content, and (iii) improved generation stability, assessed through the consistency of images generated from the refinement. This

process yields more than 11,000 original–refined prompt pairs for supervised fine-tuning.

For the information aligner, we construct more than 4,000 samples containing the user intent, original prompt, and enhancer-produced refinement. Paired images are generated with SD 2.1 (Rombach et al. 2022), CoMat 2.1 (Jiang et al. 2024), Stable Diffusion 3 (Esser et al. 2024), Flux.1 (Labs 2024), PixArt- Σ (Saharia et al. 2022), Infinity (Han et al. 2025), and Show-O2 (Xie, Yang, and Shou 2025). Qwen3-VL evaluates image–intent consistency, and the improvement from the original prompt to the refinement provides the rewrite-supervision signal. The aligner dataset is divided into training and validation sets at an 8:2 ratio.

Training Hyperparameters Settings. We trained our model with the following hyperparameters: a learning rate of 1×10^{-5} , a batch size of 2, gradient accumulation steps of 16, and a total of 3 training epochs. The checkpoint with the lowest training loss was selected as the final model.

During LoRA fine-tuning, all parameters of the base model were frozen, and only the LoRA parameters were updated, specifically for query and value (Vaswani et al. 2017; Hu et al. 2022). The LoRA hyperparameters were set as follows: rank $r = 8$, $\alpha = 16$, a dropout rate of 0.1, and no bias. Training was performed using bf16 mixed precision.

We employed the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-2} .

We train enhancer variants with Mistral-7B (Jiang et al. 2023), Llama-3.1-8B (Meta AI 2024), and Qwen3-8B (Team 2025), using Llama-3.1-8B as the default backbone. Training and evaluation use AMD MI250X GPUs and fixed random seeds.

Information Canonicalizer and Filter

During supervision construction, the information canonicalizer receives a user prompt P and its generated image X and reconstructs a semantically explicit, well-organized canonical prompt:

$$\hat{P}^* = g_{\text{can}}(P, X).$$

The image supplies visual evidence for reorganizing the information expressed by the prompt; canonicalization is therefore not an image-only inverse-prompting procedure. Its output must preserve the original intent rather than merely imitate model-specific prompt syntax.

Because a reconstructed candidate may contain visual hallucinations or semantic deviations, the information filter retains only candidates that preserve the user intent and provide valid information reorganization:

$$\mathcal{D} = \left\{ (P, \hat{P}^*) \mid g_{\text{fil}}(P, \hat{P}^*, X, Y) = 1 \right\}.$$

Here, Y is used during data construction to verify intent preservation. The filtered prompt pairs form the supervision set for the enhancer; neither X nor a separate Y annotation is required by the trained enhancer at inference time.

Figure 1 shows the image-analysis instruction component used by the information canonicalizer. The source prompt P is provided separately as contextual evidence in the canonicalization pipeline; the subsequent information filter checks

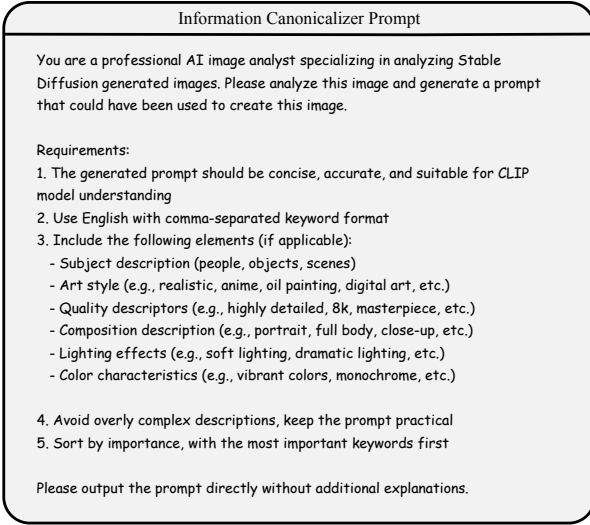


Figure 1: Information Canonicalizer prompt used during supervision construction. The template asks for a concise, structured description grounded in the supplied image; source-prompt consistency is enforced by the surrounding canonicalization and filtering pipeline.

the reconstructed candidate against both the source prompt and the stated intent before it can enter the enhancer’s supervision set.

Information Enhancer

The information enhancer learns to map an original prompt into the canonical prompt subspace,

$$P^* \sim \pi_\theta(\cdot | P),$$

using the filtered pairs in $\mathcal{D}$. Its supervised fine-tuning objective is

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(P, \hat{P}^*) \sim \mathcal{D}} [\log \pi_\theta(\hat{P}^* | P)].$$

The enhancer generates an intent-preserving candidate refinement from text alone. Whether that candidate replaces the original prompt is decided subsequently by the information aligner, rather than by unconditional prompt expansion.

The inference-time prompt template is shown in Figure 2. It prioritizes preservation of all explicit entities, attributes, relations, scenes, and styles, while prohibiting unsupported additions and meaningful omissions.

Information Aligner Implementation

The information aligner is a learned predictor that determines whether rewriting an original prompt P into P^* is expected to improve semantic alignment for a target text-to-image generator ϕ_m . Following the main-text notation, its training-time formulation is

$$a_\psi(P, P^*, \phi_m, Y) = p_\psi(z = 1 | P, P^*, \phi_m, Y),$$

where Y denotes the user intent and z indicates whether the rewrite improves the image-side Qwen3-VL alignment

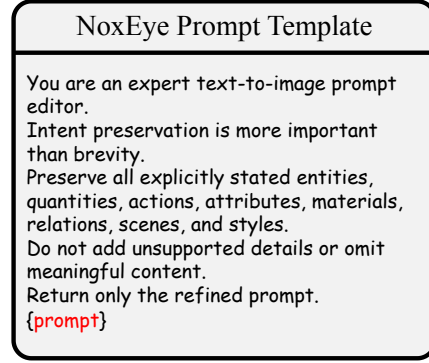


Figure 2: NoxEye prompt template for generating an intent-preserving candidate rewrite from the original prompt.

score. In this expression, Y and the outputs of ϕ_m belong to the supervision process: they are used to determine whether a prompt pair should receive a positive label. The training label is

$$z = \mathbb{I}[\bar{Q}(\phi_m(P^*), Y) - \bar{Q}(\phi_m(P), Y) > \delta].$$

Here, $\bar{Q}$ is the Qwen3-VL semantic-alignment score averaged over multiple generated images. For each original or refined prompt, we generate eight images. The original and refined prompts use the same eight random seeds so that their score difference is not confounded by unmatched sampling noise. The margin δ is obtained from the training data rather than fixed analytically.

Training-Time versus Inference-Time Inputs. The notation $a_\psi(P, P^*, \phi_m, Y)$ describes how supervision is constructed during training; it does not mean that a separately annotated intent Y or generated images must be supplied at inference time. During training, Y , the target generator ϕ_m , the eight generated images, and their Qwen3-VL scores are used only to construct z and the continuous gain target. At inference time, the deployed aligner receives only the original prompt P and its candidate rewrite P^* . Its input features are computed directly from this text pair. For a selected target generator, the corresponding text-encoder feature extractor is fixed by the scoring pipeline, so neither the generator identity nor an image is provided as an additional user input. We denote the deployed decision function as

$$a_\psi^{\text{infer}}(P, P^*) = p_\psi(z = 1 | f(P, P^*)),$$

where $f(P, P^*)$ collects the surface, Qwen embedding, and target-text-encoder features described below. Consequently, routing does not require image generation or access to a separate natural-language intent annotation at deployment time.

Text Surface Features. We organize the surface statistics into three complementary groups:

- *Length and structure:* original_token_count, rewrite_token_count, length_ratio, length_difference, character_ratio, and sentence_count_difference.

- *Lexical overlap*: token_jaccard, word_overlap_recall, and word_overlap_precision.
- *Edit and content change*: normalized_edit_distance, longest_common_subsequence_ratio, new_token_ratio, and deleted_token_ratio.

Together, these features characterize how much information is retained, added, or removed by the rewrite without treating prompt length alone as a measure of semantic quality.

Qwen Embedding Features. We directly extract text embeddings for the original and rewritten prompts using Qwen/Qwen3-Embedding-0.6B. These embeddings provide model-independent semantic features of the prompt pair; they are extracted directly rather than reconstructed from generated images.

Target T2I Text Encoder Features. Because rewrite utility depends on the target generator, target-model text embeddings are extracted separately using the text encoder associated with each text-to-image model ϕ_m . The resulting model-specific features allow the aligner to account for differences in how individual generators encode the same original and rewritten prompts.

Gain Regression and Routing. In the selector implementation, we additionally train a regressor to predict the continuous gain

$$y = \text{qwen_score}(P^*) - \text{qwen_score}(P),$$

where each score is the eight-image average defined above. We consider classical regressors including histogram-based gradient boosting (HistGBR), XGBoost, Elastic Net, and ridge regression. The data are divided into training and validation sets using an 8:2 split. At inference time, the routing rule is

$$\begin{cases} \text{choose the rewrite,} & \hat{y} > \tau, \\ \text{retain the original prompt,} & \hat{y} \leq \tau, \end{cases}$$

where the routing threshold τ is selected on the validation set. This formulation trains the aligner as a rewrite-benefit predictor rather than as a composite reward model.

Baseline Implementations

We compare against dedicated prompt-optimization methods Promptist (Hao et al. 2023), PAG (Yun et al. 2025), and PAE (Mo et al. 2024); closed-model rewriters Gemini 2.0 Flash and DeepSeek-V4-Flash; and open-weight rewriters based on Mistral-7B, Llama-3.1-8B, and Qwen3-8B. Public implementations are used when available. CoMat 2.1 is reproduced under the same experimental protocol. All methods use identical downstream generation settings for each target text-to-image model.

Generation and Evaluation Settings

For evaluation on **AP Bench**, **GenEval**, and **T2I-CompBench**, all rewriting methods are evaluated under the same generation protocol for a given target model. For SD2.1 (Rombach et al. 2022) and CoMat2.1 (Jiang et al.

Table 1: CLIP target-category probability mean ($\uparrow$) on all AP Bench samples. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.3733	0.3900	0.4340	0.4093	0.3819	0.3909	0.3444
Promptist (Hao et al. 2023)	0.3480	0.3655	0.3977	0.3751	0.3619	0.3731	0.3317
PAG (Yun et al. 2025)	0.2920	0.3019	0.3257	0.3060	0.2933	0.3033	0.2737
PAE (Mo et al. 2024)	0.3193	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.3681	0.3833	0.4249	0.4040	0.3864	0.3982	0.3621
Llama-3.1-8B (Meta AI 2024)	0.3466	0.3598	0.4006	0.3775	0.3586	0.3766	0.3337
Qwen3-8B (Team 2025)	0.3662	0.3786	0.4171	0.3948	0.3756	0.3919	0.3601
Gemini 2.0 Flash closed-source	0.3766	0.3944	0.4315	0.4085	0.3862	0.3916	0.3415
DeepSeek-V4-Flash closed-source	0.3639	0.3830	0.4258	0.4000	0.3755	0.3867	0.3502
NoxEye (Ours) w/o Aligner	0.3654	0.3818	0.4173	0.3951	0.3792	0.3908	0.3459
NoxEye (Ours)	0.3751	0.3888	0.4286	0.4079	0.3834	0.3903	0.3487

2024), the number of sampling steps is 50, the CFG scale is 7.5, and the default image size is used. For other generators, we use their default generation hyperparameters. Fixed random seeds are used to support paired and reproducible comparisons.

Full Results on AP Bench

The following tables use the same formal comparison methods and visual grouping as the main paper. Each prompt is evaluated with eight generated images. CLIP mean and standard deviation are computed over the available sample-seed target-category probabilities, and Qwen3-VL is averaged over the same images. Following correction of the source difficulty labels, Medium corresponds to the former Hard split (“intent not confirmed”) and therefore has no closed-set CLIP target category; Hard corresponds to the former Medium split (“intent confirmed from prompt only”) and reports both CLIP and Qwen3-VL. Missing PAE results are shown as “—”.

Table 2: CLIP target-category probability standard deviation ($\downarrow$) on all AP Bench samples. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.3793	0.3821	0.3812	0.3745	0.3760	0.3784	0.3640
Promptist (Hao et al. 2023)	0.3783	0.3818	0.3834	0.3777	0.3758	0.3827	0.3681
PAG (Yun et al. 2025)	0.3726	0.3749	0.3849	0.3721	0.3673	0.3774	0.3593
PAE (Mo et al. 2024)	0.3639	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.3794	0.3828	0.3804	0.3749	0.3757	0.3783	0.3648
Llama-3.1-8B (Meta AI 2024)	0.3801	0.3824	0.3851	0.3779	0.3750	0.3846	0.3664
Qwen3-8B (Team 2025)	0.3783	0.3798	0.3802	0.3753	0.3732	0.3800	0.3686
Gemini 2.0 Flash closed-source	0.3812	0.3831	0.3824	0.3774	0.3749	0.3787	0.3624
DeepSeek-V4-Flash closed-source	0.3780	0.3814	0.3814	0.3759	0.3750	0.3786	0.3637
NoxEye (Ours) w/o Aligner	0.3775	0.3815	0.3777	0.3706	0.3717	0.3774	0.3614
NoxEye (Ours)	0.3789	0.3813	0.3802	0.3748	0.3749	0.3771	0.3640

Table 3: Qwen3-VL semantic-alignment mean ($\uparrow$) on all AP Bench samples. Each non-original cell reports the absolute score and its change relative to Original Prompt. Bold and underlined values indicate the best and second-best results.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.6544	0.6591	0.7379	0.7579	0.7536	0.7182	0.6847
Promptist (Hao et al. 2023)	0.5813 (-0.0731)	0.5876 (-0.0715)	0.6412 (-0.0967)	0.6569 (-0.1010)	0.6620 (-0.0916)	0.6335 (-0.0847)	0.6062 (-0.0785)
PAG (Yun et al. 2025)	0.4174 (-0.2370)	0.4245 (-0.2346)	0.4580 (-0.2799)	0.4754 (-0.2825)	0.4817 (-0.2719)	0.4550 (-0.2632)	0.4369 (-0.2478)
PAE (Mo et al. 2024)	0.6246 (-0.0298)	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.6435 (-0.0109)	0.6473 (-0.0118)	0.7224 (-0.0155)	0.7335 (-0.0244)	0.7434 (-0.0102)	0.7237 (+0.0055)	0.6963 (+0.0116)
Llama-3.1-8B (Meta AI 2024)	0.5736 (-0.0808)	0.5803 (-0.0788)	0.6441 (-0.0938)	0.6624 (-0.0955)	0.6713 (-0.0823)	0.6499 (-0.0683)	0.6175 (-0.0672)
Qwen3-8B (Team 2025)	0.6404 (-0.0140)	0.6442 (-0.0149)	0.7204 (-0.0175)	0.7418 (-0.0161)	0.7366 (-0.0170)	0.7149 (-0.0033)	0.6884 (+0.0037)
Gemini 2.0 Flash closed-source	0.6574 (+0.0030)	0.6633 (+0.0042)	0.7201 (-0.0178)	0.7461 (-0.0118)	0.7464 (-0.0072)	0.7156 (-0.0026)	0.6801 (-0.0046)
DeepSeek-V4-Flash closed-source	0.6486 (-0.0058)	0.6534 (-0.0057)	0.7207 (-0.0172)	0.7416 (-0.0163)	0.7423 (-0.0113)	0.7133 (-0.0049)	0.6860 (+0.0013)
NoxEye (Ours) w/o Aligner	0.6493 (-0.0051)	0.6569 (-0.0022)	0.7266 (-0.0113)	0.7440 (-0.0139)	0.7512 (-0.0024)	0.7257 (+0.0075)	0.6963 (+0.0116)
NoxEye (Ours)	0.6613 (+0.0069)	0.6669 (+0.0078)	0.7412 (+0.0033)	0.7626 (+0.0047)	0.7594 (+0.0058)	0.7335 (+0.0153)	0.7014 (+0.0167)

Table 4: CLIP target-category probability mean ($\uparrow$) on Easy Novice. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.4273	0.4447	0.4936	0.4652	0.4595	0.4660	0.4081
Promptist (Hao et al. 2023)	0.3999	0.4188	0.4583	0.4296	0.4402	0.4522	0.3995
PAG (Yun et al. 2025)	0.3495	0.3625	0.3834	0.3629	0.3594	0.3740	0.3352
PAE (Mo et al. 2024)	0.3659	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.4137	0.4270	0.4821	0.4569	0.4593	0.4616	0.4245
Llama-3.1-8B (Meta AI 2024)	0.3879	0.4010	0.4470	0.4269	0.4225	0.4342	0.3931
Qwen3-8B (Team 2025)	0.4179	0.4306	0.4769	0.4476	0.4492	0.4643	0.4276
Gemini 2.0 Flash closed-source	0.4311	0.4537	0.4889	0.4648	0.4634	0.4621	0.4030
DeepSeek-V4-Flash closed-source	0.4117	0.4321	0.4810	0.4505	0.4494	0.4561	0.4130
NoxEye (Ours) w/o Aligner	0.4161	0.4324	0.4744	0.4492	0.4514	0.4565	0.4096
NoxEye (Ours)	0.4295	0.4413	0.4868	0.4625	0.4599	0.4592	0.4118

Table 5: CLIP target-category probability standard deviation ($\downarrow$) on Easy Novice. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.3721	0.3716	0.3651	0.3618	0.3679	0.3685	0.3647
Promptist (Hao et al. 2023)	0.3742	0.3754	0.3719	0.3694	0.3708	0.3758	0.3684
PAG (Yun et al. 2025)	0.3774	0.3791	0.3834	0.3737	0.3727	0.3795	0.3664
PAE (Mo et al. 2024)	0.3648	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.3748	0.3739	0.3656	0.3637	0.3664	0.3650	0.3608
Llama-3.1-8B (Meta AI 2024)	0.3774	0.3779	0.3743	0.3720	0.3721	0.3776	0.3673
Qwen3-8B (Team 2025)	0.3720	0.3711	0.3663	0.3656	0.3656	0.3701	0.3660
Gemini 2.0 Flash closed-source	0.3747	0.3741	0.3672	0.3638	0.3665	0.3709	0.3637
DeepSeek-V4-Flash closed-source	0.3710	0.3717	0.3666	0.3666	0.3684	0.3693	0.3614
NoxEye (Ours) w/o Aligner	0.3695	0.3733	0.3632	0.3592	0.3641	0.3686	0.3608
NoxEye (Ours)	0.3711	0.3710	0.3641	0.3632	0.3662	0.3678	0.3636

Table 6: Qwen3-VL semantic-alignment mean ($\uparrow$) on Easy Novice. Each non-original cell reports the absolute score and its change relative to Original Prompt. Bold and underlined values indicate the best and second-best results.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.7038	0.7069	0.7782	0.7885	0.7941	0.7577	0.7224
Promptist (Hao et al. 2023)	0.6685 (-0.0353)	0.6740 (-0.0329)	0.7435 (-0.0347)	0.7537 (-0.0348)	0.7628 (-0.0313)	0.7316 (-0.0261)	0.7079 (-0.0145)
PAG (Yun et al. 2025)	0.5198 (-0.1840)	0.5305 (-0.1764)	0.5713 (-0.2069)	0.5970 (-0.1915)	0.5996 (-0.1945)	0.5738 (-0.1839)	0.5507 (-0.1717)
PAE (Mo et al. 2024)	0.6596 (-0.0442)	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.6962 (-0.0076)	0.6958 (-0.0111)	0.7795 (+0.0013)	0.7801 (-0.0084)	0.7865 (-0.0076)	0.7779 (+0.0202)	0.7498 (+0.0274)
Llama-3.1-8B (Meta AI 2024)	0.6156 (-0.0882)	0.6219 (-0.0850)	0.6938 (-0.0844)	0.7056 (-0.0829)	0.7166 (-0.0775)	0.7003 (-0.0574)	0.6602 (-0.0622)
Qwen3-8B (Team 2025)	0.6992 (-0.0046)	0.7055 (-0.0014)	0.7763 (-0.0019)	0.7839 (-0.0046)	0.7841 (-0.0100)	0.7672 (+0.0095)	<u>0.7432</u> (+0.0208)
Gemini 2.0 Flash closed-source	0.6998 (-0.0040)	0.7035 (-0.0034)	0.7605 (-0.0177)	0.7825 (-0.0060)	0.7884 (-0.0057)	0.7629 (+0.0052)	0.7151 (-0.0073)
DeepSeek-V4-Flash closed-source	0.6974 (-0.0064)	0.7016 (-0.0053)	0.7572 (-0.0210)	0.7667 (-0.0218)	0.7786 (-0.0155)	0.7526 (-0.0051)	0.7236 (+0.0012)
NoxEye (Ours) w/o Aligner	0.6942 (-0.0096)	0.7025 (-0.0044)	0.7736 (-0.0046)	0.7753 (-0.0132)	0.7872 (-0.0069)	0.7675 (+0.0098)	0.7355 (+0.0131)
NoxEye (Ours)	0.7054 (+0.0016)	0.7115 (+0.0046)	0.7808 (+0.0026)	0.7920 (+0.0035)	0.7940 (+0.0001)	<u>0.7718</u> (+0.0141)	0.7361 (+0.0137)

Table 7: CLIP target-category probability mean ($\uparrow$) on Easy Expert. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.3753	0.3972	0.4368	0.4223	0.3643	0.3795	0.3291
Promptist (Hao et al. 2023)	0.3450	0.3651	0.3847	0.3778	0.3260	0.3483	0.3011
PAG (Yun et al. 2025)	0.2754	0.2819	0.3120	0.2890	0.2607	0.2710	0.2397
PAE (Mo et al. 2024)	0.3211	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.3819	0.4035	0.4284	0.4166	0.3664	0.3987	0.3489
Llama-3.1-8B (Meta AI 2024)	0.3702	0.3879	0.4265	0.4018	0.3608	0.3880	0.3227
Qwen3-8B (Team 2025)	0.3704	0.3866	0.4155	0.4081	0.3572	0.3799	0.3402
Gemini 2.0 Flash closed-source	0.3882	0.4022	0.4383	0.4196	0.3692	0.3828	0.3292
DeepSeek-V4-Flash closed-source	0.3738	0.3968	0.4331	0.4180	0.3617	0.3782	0.3346
NoxEye (Ours) w/o Aligner	0.3744	0.3951	0.4215	0.4067	0.3677	0.3874	0.3299
NoxEye (Ours)	0.3774	0.3992	0.4341	0.4212	0.3677	0.3861	0.3326

Table 8: CLIP target-category probability standard deviation ($\downarrow$) on Easy Expert. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.3856	0.3914	0.3878	0.3815	0.3761	0.3818	0.3596
Promptist (Hao et al. 2023)	0.3867	0.3922	0.3928	0.3878	0.3740	0.3846	0.3669
PAG (Yun et al. 2025)	0.3791	0.3801	0.3959	0.3793	0.3653	0.3816	0.3565
PAE (Mo et al. 2024)	0.3686	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.3863	0.3935	0.3887	0.3814	0.3778	0.3877	0.3664
Llama-3.1-8B (Meta AI 2024)	0.3886	0.3924	0.3945	0.3850	0.3787	0.3909	0.3656
Qwen3-8B (Team 2025)	0.3866	0.3888	0.3868	0.3826	0.3744	0.3834	0.3659
Gemini 2.0 Flash closed-source	0.3899	0.3900	0.3894	0.3862	0.3755	0.3797	0.3593
DeepSeek-V4-Flash closed-source	0.3863	0.3919	0.3879	0.3818	0.3755	0.3826	0.3622
NoxEye (Ours) w/o Aligner	0.3885	0.3907	0.3864	0.3781	0.3744	0.3808	0.3588
NoxEye (Ours)	0.3862	0.3907	0.3880	0.3812	0.3750	0.3810	0.3612

Table 9: Qwen3-VL semantic-alignment mean ($\uparrow$) on Easy Expert. Each non-original cell reports the absolute score and its change relative to Original Prompt. Bold and underlined values indicate the best and second-best results.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.6323	0.6374	0.7290	0.7589	0.7321	0.6971	0.6711
Promptist (Hao et al. 2023)	0.5237 (-0.1086)	0.5317 (-0.1057)	0.5759 (-0.1531)	0.6003 (-0.1586)	0.5954 (-0.1367)	0.5655 (-0.1316)	0.5372 (-0.1339)
PAG (Yun et al. 2025)	0.3462 (-0.2861)	0.3546 (-0.2828)	0.3875 (-0.3415)	0.3916 (-0.3673)	0.4023 (-0.3298)	0.3722 (-0.3249)	0.3579 (-0.3132)
PAE (Mo et al. 2024)	0.6167 (-0.0156)	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.6159 (-0.0164)	0.6184 (-0.0190)	0.6999 (-0.0291)	0.7198 (-0.0391)	0.7194 (-0.0127)	0.6919 (-0.0052)	0.6747 (+0.0036)
Llama-3.1-8B (Meta AI 2024)	0.5659 (-0.0664)	0.5735 (-0.0639)	0.6313 (-0.0977)	0.6593 (-0.0996)	0.6613 (-0.0708)	0.6331 (-0.0640)	0.6056 (-0.0655)
Qwen3-8B (Team 2025)	0.6133 (-0.0190)	0.6107 (-0.0267)	0.6969 (-0.0321)	0.7268 (-0.0321)	0.7131 (-0.0190)	0.6854 (-0.0117)	0.6630 (-0.0081)
Gemini 2.0 Flash closed-source	0.6355 (+0.0032)	0.6443 (+0.0069)	0.7097 (-0.0193)	0.7308 (-0.0281)	0.7291 (-0.0030)	0.6782 (-0.0189)	0.6669 (-0.0042)
DeepSeek-V4-Flash closed-source	0.6274 (-0.0049)	0.6325 (-0.0049)	0.7146 (-0.0144)	0.7443 (-0.0146)	0.7243 (-0.0078)	0.6874 (-0.0097)	0.6704 (-0.0007)
NoxEye (Ours) w/o Aligner	0.6348 (+0.0025)	0.6463 (+0.0089)	0.7164 (-0.0126)	0.7435 (-0.0154)	0.7458 (+0.0137)	0.7122 (+0.0151)	0.6949 (+0.0238)
NoxEye (Ours)	0.6533 (+0.0210)	0.6471 (+0.0097)	0.7335 (+0.0045)	0.7607 (+0.0018)	0.7404 (+0.0083)	0.7172 (+0.0201)	0.6938 (+0.0227)

Table 10: Qwen3-VL semantic-alignment mean ($\uparrow$) on Medium Novice. Each non-original cell reports the absolute score and its change relative to Original Prompt. Bold and underlined values indicate the best and second-best results.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.6321	0.6251	0.6955	0.7090	0.7189	0.6820	0.6347
Promptist (Hao et al. 2023)	0.6416 (+0.0095)	0.6302 (+0.0051)	0.7035 (+0.0080)	0.7071 (-0.0019)	0.7222 (+0.0033)	0.7013 (+0.0193)	0.6282 (-0.0065)
PAG (Yun et al. 2025)	0.5222 (-0.1099)	0.5069 (-0.1182)	0.5074 (-0.1881)	0.5567 (-0.1523)	0.5521 (-0.1668)	0.5586 (-0.1234)	0.4772 (-0.1575)
PAE (Mo et al. 2024)	0.5885 (-0.0436)	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.5944 (-0.0377)	0.6078 (-0.0173)	0.6624 (-0.0331)	0.6669 (-0.0421)	0.7081 (-0.0108)	0.6817 (-0.0003)	0.6534 (+0.0187)
Llama-3.1-8B (Meta AI 2024)	0.5828 (-0.0493)	0.6063 (-0.0188)	0.6479 (-0.0476)	0.6797 (-0.0293)	0.7042 (-0.0147)	0.6787 (-0.0033)	0.6449 (+0.0102)
Qwen3-8B (Team 2025)	0.6134 (-0.0187)	0.6161 (-0.0090)	0.6807 (-0.0148)	0.7137 (+0.0047)	0.7174 (-0.0015)	0.6985 (+0.0165)	0.6459 (+0.0112)
Gemini 2.0 Flash closed-source	0.6029 (-0.0292)	0.6185 (-0.0066)	0.7007 (+0.0052)	0.7037 (-0.0053)	0.7284 (+0.0095)	0.6913 (+0.0093)	0.6247 (-0.0100)
DeepSeek-V4-Flash closed-source	0.6258 (-0.0063)	0.6222 (-0.0029)	0.6820 (-0.0135)	0.7006 (-0.0084)	0.7031 (-0.0158)	0.6882 (+0.0062)	0.6013 (-0.0334)
NoxEye (Ours) w/o Aligner	0.6216 (-0.0105)	0.6471 (+0.0220)	0.6876 (-0.0079)	0.7041 (-0.0049)	0.7341 (+0.0152)	0.7381 (+0.0561)	0.6683 (+0.0336)
NoxEye (Ours)	0.6539 (+0.0218)	0.6390 (+0.0139)	0.6991 (+0.0036)	0.7250 (+0.0160)	0.7346 (+0.0157)	0.7305 (+0.0485)	0.6641 (+0.0294)

Table 11: Qwen3-VL semantic-alignment mean ($\uparrow$) on Medium Expert. Each non-original cell reports the absolute score and its change relative to Original Prompt. Bold and underlined values indicate the best and second-best results.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.6460	0.6615	0.7095	0.7313	0.7309	0.6729	0.6488
Promptist (Hao et al. 2023)	0.5012 (-0.1448)	0.5139 (-0.1476)	0.5304 (-0.1791)	0.5388 (-0.1925)	0.5541 (-0.1768)	0.5221 (-0.1508)	0.5053 (-0.1435)
PAG (Yun et al. 2025)	0.3248 (-0.3212)	0.3293 (-0.3322)	0.3448 (-0.3647)	0.3611 (-0.3702)	0.3833 (-0.3476)	0.3337 (-0.3392)	0.3503 (-0.2985)
PAE (Mo et al. 2024)	0.6165 (-0.0295)	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.6318 (-0.0142)	0.6497 (-0.0118)	0.6999 (-0.0096)	0.7183 (-0.0130)	0.7519 (+0.0210)	0.7127 (+0.0398)	0.6645 (+0.0157)
Llama-3.1-8B (Meta AI 2024)	0.5570 (-0.0890)	0.5659 (-0.0956)	0.6141 (-0.0954)	0.6149 (-0.1164)	0.6163 (-0.1146)	0.6033 (-0.0696)	0.5607 (-0.0881)
Qwen3-8B (Team 2025)	0.6127 (-0.0333)	0.6190 (-0.0425)	0.6855 (-0.0240)	0.7227 (-0.0086)	0.7048 (-0.0261)	0.6730 (+0.0001)	0.6541 (+0.0053)
Gemini 2.0 Flash closed-source	0.6435 (-0.0025)	0.6488 (-0.0127)	0.6968 (-0.0127)	0.7246 (-0.0067)	0.7060 (-0.0249)	0.6793 (+0.0064)	0.6527 (+0.0039)
DeepSeek-V4-Flash closed-source	0.6392 (-0.0068)	0.6510 (-0.0105)	0.6901 (-0.0194)	0.7206 (-0.0107)	0.7263 (-0.0046)	0.6720 (-0.0009)	0.6618 (+0.0130)
NoxEye (Ours) w/o Aligner	0.6414 (-0.0046)	0.6422 (-0.0193)	0.7069 (-0.0026)	0.7377 (+0.0064)	0.7366 (+0.0057)	0.6938 (+0.0209)	0.6756 (+0.0268)
NoxEye (Ours)	0.6435 (-0.0025)	0.6798 (+0.0183)	0.7255 (+0.0160)	0.7479 (+0.0166)	0.7568 (+0.0259)	0.7013 (+0.0284)	0.6845 (+0.0357)

Table 12: CLIP target-category probability mean ($\uparrow$) on Hard Novice. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.1450	0.1380	0.1870	0.2145	0.1628	0.1696	0.1632
Promptist (Hao et al. 2023)	0.1356	0.1209	0.1921	0.1882	0.1772	0.1690	0.1647
PAG (Yun et al. 2025)	0.1194	0.1213	0.1774	0.1925	0.1520	0.1645	0.1499
PAE (Mo et al. 2024)	0.1232	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.1521	0.1474	0.1884	0.1979	0.1587	0.1809	0.1615
Llama-3.1-8B (Meta AI 2024)	0.1412	0.1382	0.1834	0.1757	0.1338	0.1902	0.1484
Qwen3-8B (Team 2025)	0.1233	0.1295	0.2014	0.2086	0.1492	0.1885	0.1943
Gemini 2.0 Flash closed-source	0.1464	0.1409	0.1870	0.1961	0.1572	0.1510	0.1434
DeepSeek-V4-Flash closed-source	0.1515	0.1533	0.1859	0.1876	0.1638	0.1810	0.1797
NoxEye (Ours) w/o Aligner	0.1465	0.1149	0.1938	0.2039	0.1676	0.1888	0.2002
NoxEye (Ours)	0.1429	0.1343	0.1854	0.2140	0.1724	0.1845	0.1900

Table 13: CLIP target-category probability standard deviation ($\downarrow$) on Hard Novice. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.2966	0.2964	0.3195	0.3482	0.3150	0.3106	0.3060
Promptist (Hao et al. 2023)	0.2848	0.2667	0.3242	0.3263	0.3279	0.3083	0.3141
PAG (Yun et al. 2025)	0.2603	0.2670	0.3125	0.3299	0.2993	0.3031	0.2973
PAE (Mo et al. 2024)	0.2660	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.3012	0.3021	0.3131	0.3251	0.3034	0.3207	0.2950
Llama-3.1-8B (Meta AI 2024)	0.3078	0.2972	0.3447	0.3268	0.2762	0.3517	0.3020
Qwen3-8B (Team 2025)	0.2535	0.2816	0.3284	0.3358	0.2962	0.3290	0.3348
Gemini 2.0 Flash closed-source	0.2893	0.2818	0.3114	0.3272	0.2960	0.2947	0.2961
DeepSeek-V4-Flash closed-source	0.2986	0.3027	0.3130	0.3149	0.3159	0.3140	0.3221
NoxEye (Ours) w/o Aligner	0.3020	0.2650	0.3337	0.3451	0.3188	0.3429	0.3635
NoxEye (Ours)	0.2953	0.2964	0.3231	0.3484	0.3285	0.3389	0.3417

Table 14: Qwen3-VL semantic-alignment mean ($\uparrow$) on Hard Novice. Each non-original cell reports the absolute score and its change relative to Original Prompt. Bold and underlined values indicate the best and second-best results.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.6692	0.6667	0.7500	0.7787	0.7134	0.7113	0.6572
Promptist (Hao et al. 2023)	0.6336 (-0.0356)	0.6516 (-0.0151)	0.6845 (-0.0655)	0.7185 (-0.0602)	0.6745 (-0.0389)	0.6523 (-0.0590)	0.6336 (-0.0236)
PAG (Yun et al. 2025)	0.3778 (-0.2914)	0.3660 (-0.3007)	0.3951 (-0.3549)	0.4854 (-0.2933)	0.4438 (-0.2696)	0.3993 (-0.3120)	0.3850 (-0.2722)
PAE (Mo et al. 2024)	0.6576 (-0.0116)	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.6556 (-0.0136)	0.6667 (+0.0000)	0.7259 (-0.0241)	0.7340 (-0.0447)	0.7345 (+0.0211)	0.7634 (+0.0521)	0.6752 (+0.0180)
Llama-3.1-8B (Meta AI 2024)	0.5407 (-0.1285)	0.5333 (-0.1334)	0.5664 (-0.1836)	0.5801 (-0.1986)	0.5271 (-0.1863)	0.5032 (-0.2081)	0.4988 (-0.1584)
Qwen3-8B (Team 2025)	0.6611 (-0.0081)	0.6486 (-0.0181)	0.6975 (-0.0525)	0.7537 (-0.0250)	0.7326 (+0.0192)	0.7167 (+0.0054)	0.6183 (-0.0389)
Gemini 2.0 Flash closed-source	0.6588 (-0.0104)	0.6639 (-0.0028)	0.7384 (-0.0116)	0.7898 (+0.0111)	0.7056 (-0.0078)	0.7426 (+0.0313)	0.6625 (+0.0053)
DeepSeek-V4-Flash closed-source	0.6924 (+0.0232)	0.7035 (+0.0368)	0.7282 (-0.0218)	0.7970 (+0.0183)	0.7081 (-0.0053)	0.7431 (+0.0318)	0.6613 (+0.0041)
NoxEye (Ours) w/o Aligner	0.6574 (-0.0118)	0.6481 (-0.0186)	0.6771 (-0.0729)	0.7324 (-0.0463)	0.7021 (-0.0113)	0.7127 (+0.0014)	0.5977 (-0.0595)
NoxEye (Ours)	0.6866 (+0.0174)	0.6479 (-0.0188)	0.7701 (+0.0201)	0.8123 (+0.0336)	0.6919 (-0.0215)	0.7225 (+0.0112)	0.6583 (+0.0011)

Table 15: CLIP target-category probability mean ($\uparrow$) on Hard Expert. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.1460	0.1484	0.1806	0.1393	0.0969	0.1010	0.1130
Promptist (Hao et al. 2023)	0.1424	0.1494	0.1774	0.1389	0.1191	0.0991	0.1206
PAG (Yun et al. 2025)	0.0918	0.0978	0.1137	0.1021	0.0927	0.0804	0.0955
PAE (Mo et al. 2024)	0.1228	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.1462	0.1550	0.1800	0.1497	0.1309	0.1304	0.1332
Llama-3.1-8B (Meta AI 2024)	0.1168	0.1206	0.1421	0.1074	0.0847	0.1014	0.1109
Qwen3-8B (Team 2025)	0.1453	0.1469	0.1713	0.1367	0.1120	0.1141	0.1213
Gemini 2.0 Flash closed-source	0.1215	0.1299	0.1772	0.1436	0.1026	0.1183	0.1146
DeepSeek-V4-Flash closed-source	0.1421	0.1467	0.1798	0.1432	0.0966	0.1136	0.1224
NoxEye (Ours) w/o Aligner	0.1329	0.1445	0.1687	0.1362	0.1015	0.1207	0.1119
NoxEye (Ours)	0.1461	0.1487	0.1743	0.1428	0.0972	0.1073	0.1199

Table 16: CLIP target-category probability standard deviation ($\downarrow$) on Hard Expert. Absolute probabilities are reported without rank-based emphasis.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.3013	0.3001	0.3237	0.2809	0.2423	0.2451	0.2575
Promptist (Hao et al. 2023)	0.2953	0.2973	0.3184	0.2847	0.2663	0.2499	0.2642
PAG (Yun et al. 2025)	0.2403	0.2460	0.2699	0.2517	0.2449	0.2337	0.2478
PAE (Mo et al. 2024)	0.2729	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.2958	0.3052	0.3203	0.2942	0.2789	0.2794	0.2700
Llama-3.1-8B (Meta AI 2024)	0.2743	0.2746	0.2929	0.2488	0.2266	0.2579	0.2592
Qwen3-8B (Team 2025)	0.2979	0.2996	0.3180	0.2792	0.2598	0.2637	0.2653
Gemini 2.0 Flash closed-source	0.2686	0.2788	0.3214	0.2908	0.2441	0.2659	0.2536
DeepSeek-V4-Flash closed-source	0.3016	0.2999	0.3279	0.2868	0.2418	0.2657	0.2678
NoxEye (Ours) w/o Aligner	0.2865	0.2944	0.3088	0.2782	0.2365	0.2664	0.2460
NoxEye (Ours)	0.2993	0.3001	0.3181	0.2827	0.2431	0.2499	0.2582

Table 17: Qwen3-VL semantic-alignment mean ($\uparrow$) on Hard Expert. Each non-original cell reports the absolute score and its change relative to Original Prompt. Bold and underlined values indicate the best and second-best results.

Method	SD2.1	CoMat	SD3	Flux.1	PixArt- Σ	Inf	Show-O2
Original Prompt	0.4978	0.5028	0.6121	0.6496	0.6596	0.6445	0.5982
Promptist (Hao et al. 2023)	0.3786 (-0.1192)	0.3826 (-0.1202)	0.4179 (-0.1942)	0.4437 (-0.2059)	0.4504 (-0.2092)	0.4392 (-0.2053)	0.4009 (-0.1973)
PAG (Yun et al. 2025)	0.1900 (-0.3078)	0.1886 (-0.3142)	0.2174 (-0.3947)	0.2140 (-0.4356)	0.2211 (-0.4385)	0.2113 (-0.4332)	0.1942 (-0.4040)
PAE (Mo et al. 2024)	0.4978 (+0.0000)	—	—	—	—	—	—
Mistral-7B (Jiang et al. 2023)	0.4977 (-0.0001)	0.5075 (+0.0047)	0.5571 (-0.0550)	0.5881 (-0.0615)	0.6131 (-0.0465)	0.5754 (-0.0691)	0.5508 (-0.0474)
Llama-3.1-8B (Meta AI 2024)	0.4157 (-0.0821)	0.4166 (-0.0862)	0.4832 (-0.1289)	0.5182 (-0.1314)	0.5455 (-0.1141)	0.5131 (-0.1314)	0.5087 (-0.0895)
Qwen3-8B (Team 2025)	0.4709 (-0.0269)	0.4812 (-0.0216)	0.5707 (-0.0414)	0.6112 (-0.0384)	0.6150 (-0.0446)	0.5947 (-0.0498)	0.5562 (-0.0420)
Gemini 2.0 Flash closed-source	0.5494 (+0.0516)	0.5550 (+0.0522)	0.5849 (-0.0272)	0.6450 (-0.0046)	0.6453 (-0.0143)	0.6339 (-0.0106)	0.5975 (-0.0007)
DeepSeek-V4-Flash closed-source	0.4887 (-0.0091)	0.4920 (-0.0108)	0.6065 (-0.0156)	0.6420 (-0.0076)	0.6538 (-0.0058)	0.6422 (-0.0023)	0.6045 (+0.0063)
NoxEye (Ours) w/o Aligner	0.4941 (-0.0037)	0.4913 (-0.0115)	0.5725 (-0.0396)	0.6198 (-0.0298)	0.6235 (-0.0361)	0.5942 (-0.0503)	0.5577 (-0.0405)
NoxEye (Ours)	0.4936 (-0.0042)	<u>0.5130</u> (+0.0102)	0.6015 (-0.0106)	0.6500 (+0.0004)	0.6692 (+0.0096)	0.6315 (-0.0130)	0.5937 (-0.0045)

Full Results on GenEval

The tables below report the current formal GenEval results for all seven evaluated backbones: SD2.1, CoMat 2.1, SD3, Flux.1, PixArt- Σ , Infinity, and Show-o2. For each method and backbone, Overall is the arithmetic mean of the six GenEval task scores. The aggregate table in the main paper first computes these scores per backbone and then takes an unweighted macro-average over the seven evaluated T2I backbones.

The full results give a more nuanced picture than the previous four-model summary. The selector-based NoxEye obtains the highest Overall score on SD2.1, CoMat 2.1, and Infinity. On SD3, Flux.1, PixArt- Σ , and Show-o2, the always-rewrite variant or another baseline is stronger, showing that rewriting gains remain model dependent. When macro-averaged over the seven backbones, NoxEye w/o Aligner and NoxEye achieve Overall scores of 0.59990 and 0.59975, respectively, compared with 0.58952 for the original prompts. NoxEye has the best macro-averaged Two Object (0.70853), Colors (0.80470), and Color Attribute (0.37464) scores, whereas NoxEye w/o Aligner has the best Counting (0.50983) and Position (0.26143) scores. Thus, GenEval does not show an aggregate advantage of the selector over always rewriting. Instead, these results show that selective rewriting preserves the competitive performance of always-rewrite strategies while providing additional protection against harmful refinements on authentic prompts, as evaluated separately on AP Bench.

Method	Overall	Single Object	Two Objects	Counting	Colors	Color Attribute	Position
Original Prompt	0.4911	0.9750	0.5101	0.4188	0.8351	0.1400	0.0675
Promptist (Hao et al. 2023)	0.4752	0.9594	0.4848	0.3781	0.8138	0.1375	0.0775
PAG (Yun et al. 2025)	0.4406	0.9500	0.4621	0.3344	0.7394	0.1150	0.0425
PAE (Mo et al. 2024)	0.3858	0.8562	0.3914	0.2312	0.6809	0.0650	0.0900
NoxEye (Ours) w/o Aligner	0.4671	0.9656	0.3939	0.4250	0.7979	0.1075	0.1125
NoxEye (Ours)	0.4966	0.9812	0.5101	0.4156	0.8404	0.1400	0.0925

Table 18: Full GenEval results on SD2.1. Bold values indicate the best results.

Method	Overall	Single Object	Two Objects	Counting	Colors	Color Attribute	Position
Original Prompt	0.4937	0.9875	0.5480	0.4219	0.7926	0.1375	0.0750
Promptist (Hao et al. 2023)	0.4803	0.9594	0.4899	0.3969	0.8059	0.1425	0.0875
PAG (Yun et al. 2025)	0.4513	0.9750	0.4066	0.3750	0.7739	0.1275	0.0500
PAE (Mo et al. 2024)	—	—	—	—	—	—	—
NoxEye (Ours) w/o Aligner	0.4836	0.9562	0.4773	0.4125	0.7979	0.1525	0.1050
NoxEye (Ours)	0.4963	0.9781	0.5177	0.4406	0.8112	0.1475	0.0825

Table 19: Full GenEval results on CoMat 2.1. Bold values indicate the best results.

Method	Overall	Single Object	Two Objects	Counting	Colors	Color Attribute	Position
Original Prompt	0.6856	0.9969	0.8561	0.5656	0.8298	0.5750	0.2900
Promptist (Hao et al. 2023)	0.6817	0.9969	0.8662	0.5562	0.8511	0.5350	0.2850
PAG (Yun et al. 2025)	0.6484	0.9906	0.8535	0.5125	0.8590	0.4975	0.1775
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.6937	0.9938	0.8712	0.5719	0.8351	0.5475	0.3425
NoxEye (Ours)	0.6849	0.9969	0.8788	0.5469	0.8271	0.5675	0.2925

Table 20: Full GenEval results on SD3. Bold values indicate the best results.

Method	Overall	Single Object	Two Objects	Counting	Colors	Color Attribute	Position
Original Prompt	0.6466	1.0000	0.8712	0.5562	0.7447	0.4575	0.2500
Promptist (Hao et al. 2023)	0.6335	0.9562	0.8182	0.5188	0.7527	0.4675	0.2875
PAG (Yun et al. 2025)	0.6108	0.9656	0.7955	0.4688	0.7952	0.4050	0.2350
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.6701	0.9688	0.8232	0.5594	0.7739	0.4675	0.4275
NoxEye (Ours)	0.6479	1.0000	0.8712	0.5562	0.7447	0.4575	0.2575

Table 21: Full GenEval results on Flux.1. Bold values indicate the best results.

Method	Overall	Single Object	Two Objects	Counting	Colors	Color Attribute	Position
Original Prompt	0.5495	0.9844	0.6389	0.4625	0.8112	0.2750	0.1250
Promptist (Hao et al. 2023)	0.5250	0.9656	0.5909	0.4406	0.7952	0.2425	0.1150
PAG (Yun et al. 2025)	0.5305	0.9719	0.6338	0.4219	0.7952	0.2425	0.1175
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.5614	0.9625	0.6641	0.4625	0.8218	0.2800	0.1775
NoxEye (Ours)	0.5538	0.9844	0.6465	0.4656	0.8138	0.2925	0.1200

Table 22: Full GenEval results on PixArt- Σ . Bold values indicate the best results.

Method	Overall	Single Object	Two Objects	Counting	Colors	Color Attribute	Position
Original Prompt	0.6742	1.0000	0.7980	0.6156	0.8191	0.5575	0.2550
Promptist (Hao et al. 2023)	0.6693	1.0000	0.7828	0.6281	0.8298	0.5425	0.2325
PAG (Yun et al. 2025)	0.6340	0.9750	0.7172	0.5312	0.8431	0.4650	0.2725
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.6751	1.0000	0.7753	0.5906	0.8324	0.5600	0.2925
NoxEye (Ours)	0.6789	1.0000	0.7677	0.6188	0.8271	0.5800	0.2800

Table 23: Full GenEval results on Infinity. Bold values indicate the best results.

Method	Overall	Single Object	Two Objects	Counting	Colors	Color Attribute	Position
Original Prompt	0.5859	0.9688	0.6717	0.5062	0.7739	0.3625	0.2325
Promptist (Hao et al. 2023)	0.6444	0.9438	0.7955	0.5437	0.7686	0.4825	0.3325
PAG (Yun et al. 2025)	0.6475	0.9688	0.7803	0.5250	0.8112	0.4975	0.3025
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.6484	0.9719	0.7753	0.5469	0.7713	0.4525	0.3725
NoxEye (Ours)	0.6398	0.9688	0.7677	0.5188	0.7686	0.4375	0.3775

Table 24: Full GenEval results on Show-o2. Bold values indicate the best results.

Full Results on T2I-CompBench

We evaluate seven backbones: Stable Diffusion 2.1 (SD2.1), CoMat 2.1, Stable Diffusion 3 (SD3), Flux.1, PixArt- Σ , Infinity, and Show-o2. Each model–method pair is evaluated on the same 11 official category–metric cells. Attribute Binding is the arithmetic mean of the BLIP scores for color, shape, texture, and complex prompts; Non-Spatial is the mean of the non-spatial and complex CLIP scores; and 2D Spatial is the mean of the spatial and complex 2D scores. The 3D Spatial, Numeracy, and 3-in-1 columns use their corresponding official scores directly. Overall is the unweighted mean of all 11 cells (four BLIP, two CLIP, two 2D, one 3D, one numeracy, and one 3-in-1 score), rather than the mean of the six displayed metric groups.

Table 5 in the main paper reports a model-level macro-average: after the above quantities are computed separately for each backbone, each column is averaged with equal weight across the seven backbones. Thus it is not weighted by the number of generated images. PAE is restricted to its supported Stable-Diffusion pipeline and no complete T2I-CompBench evaluation artifact is available under this protocol; we therefore report missing entries and exclude PAE from every macro-average, rather than imputing values or averaging it over a different model subset. “NoxEye w/o Aligner” is the always-rewrite output of the Information Enhancer, whereas NoxEye applies the learned selector between the original and enhanced prompts. All available scores are reported below; bold denotes the best available result in each column.

Method	Overall	Attribute Binding	Non Spatial	2D Spatial	3D Spatial	Numeracy	3-in-1
Original Prompt	0.3402	0.4563	0.3010	0.0924	0.3188	0.4953	0.3160
Promptist (Hao et al. 2023)	0.3121	0.4169	0.2881	0.0753	0.2948	0.4544	0.2898
PAG (Yun et al. 2025)	0.2806	0.3546	0.2885	0.0595	0.2813	0.4363	0.2548
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.3233	0.4247	0.2957	0.0817	0.3116	0.4791	0.3116
NoxEye (Ours)	0.3291	0.4353	0.2992	0.0830	0.3175	0.4851	0.3122

Table 25: Full T2I-CompBench results on SD2.1. Bold values indicate the best results.

Method	Overall	Attribute Binding	Non Spatial	2D Spatial	3D Spatial	Numeracy	3-in-1
Original Prompt	0.3523	0.4815	0.3018	0.0969	0.3312	0.4967	0.3236
Promptist (Hao et al. 2023)	0.3205	0.4324	0.2891	0.0788	0.3054	0.4625	0.2916
PAG (Yun et al. 2025)	0.2946	0.3846	0.2894	0.0602	0.2908	0.4479	0.2645
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.3346	0.4475	0.2964	0.0848	0.3225	0.4860	0.3194
NoxEye (Ours)	0.3449	0.4663	0.2999	0.0937	0.3290	0.4918	0.3208

Table 26: Full T2I-CompBench results on CoMat 2.1. Bold values indicate the best results.

Method	Overall	Attribute Binding	Non Spatial	2D Spatial	3D Spatial	Numeracy	3-in-1
Original Prompt	0.4543	0.6615	0.3049	0.1737	0.4068	0.6101	0.3770
Promptist (Hao et al. 2023)	0.4331	0.6267	0.2960	0.1619	0.3910	0.5894	0.3615
PAG (Yun et al. 2025)	0.4164	0.6018	0.2937	0.1497	0.3875	0.5613	0.3377
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.4482	0.6489	0.3025	0.1641	0.3867	0.6332	0.3813
NoxEye (Ours)	0.4530	0.6575	0.3044	0.1700	0.4053	0.6188	0.3805

Table 27: Full T2I-CompBench results on SD3. Bold values indicate the best results.

Method	Overall	Attribute Binding	Non Spatial	2D Spatial	3D Spatial	Numeracy	3-in-1
Original Prompt	0.4273	0.6220	0.3010	0.1362	0.3669	0.6067	0.3642
Promptist (Hao et al. 2023)	0.4081	0.6044	0.2944	0.1236	0.3195	0.5615	0.3541
PAG (Yun et al. 2025)	0.4051	0.5915	0.2944	0.1346	0.3471	0.5504	0.3348
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.4240	0.6168	0.2996	0.1309	0.3496	0.6174	0.3690
NoxEye (Ours)	0.4258	0.6190	0.3007	0.1354	0.3669	0.6047	0.3643

Table 28: Full T2I-CompBench results on Flux.1. Bold values indicate the best results.

Method	Overall	Attribute Binding	Non Spatial	2D Spatial	3D Spatial	Numeracy	3-in-1
Original Prompt	0.3856	0.5268	0.3013	0.1314	0.3669	0.5265	0.3754
Promptist (Hao et al. 2023)	0.3635	0.4884	0.2952	0.1181	0.3456	0.5144	0.3580
PAG (Yun et al. 2025)	0.3577	0.4872	0.2934	0.1061	0.3435	0.5110	0.3320
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.3986	0.5544	0.2998	0.1280	0.3719	0.5600	0.3792
NoxEye (Ours)	0.3904	0.5374	0.3011	0.1294	0.3709	0.5380	0.3750

Table 29: Full T2I-CompBench results on PixArt- Σ . Bold values indicate the best results.

Method	Overall	Attribute Binding	Non Spatial	2D Spatial	3D Spatial	Numeracy	3-in-1
Original Prompt	0.4155	0.5896	0.3021	0.1287	0.4063	0.5654	0.3786
Promptist (Hao et al. 2023)	0.3954	0.5531	0.2980	0.1259	0.3802	0.5523	0.3569
PAG (Yun et al. 2025)	0.3832	0.5353	0.2946	0.1232	0.3703	0.5317	0.3363
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.4266	0.6098	0.3010	0.1375	0.3866	0.6003	0.3899
NoxEye (Ours)	0.4258	0.6065	0.3011	0.1401	0.3885	0.6002	0.3865

Table 30: Full T2I-CompBench results on Infinity. Bold values indicate the best results.

Method	Overall	Attribute Binding	Non Spatial	2D Spatial	3D Spatial	Numeracy	3-in-1
Original Prompt	0.4078	0.5699	0.3035	0.1435	0.3681	0.5688	0.3756
Promptist (Hao et al. 2023)	0.4252	0.6070	0.3000	0.1440	0.4036	0.5969	0.3609
PAG (Yun et al. 2025)	0.4251	0.6043	0.2962	0.1546	0.4161	0.6040	0.3373
PAE (Mo et al. 2024)	–	–	–	–	–	–	–
NoxEye (Ours) w/o Aligner	0.4259	0.6022	0.3007	0.1535	0.3956	0.5974	0.3751
NoxEye (Ours)	0.4243	0.5990	0.3010	0.1517	0.3915	0.5978	0.3764

Table 31: Full T2I-CompBench results on Show-o2. Bold values indicate the best results.

Additional Ablations and Sensitivity Analysis

The complete component ablation is reported in the main paper; here we provide additional interpretation of the observed trends. Applying the enhancer alone does not yield uniform gains: it slightly improves CoMat 2.1 and Show-O2, but reduces alignment on SD2.1, SD3, Flux.1, PixArt- Σ , and Infinity. This confirms that a fluent, information-rich rewrite is not automatically better for every target model. Adding the information filter recovers part of the loss on SD3, Flux.1, PixArt- Σ , Infinity, and Show-O2 by removing unfaithful or weak canonicalization targets during supervision construction. However, filtering improves the quality of the enhancer’s candidate rather than deciding whether that candidate should replace a particular original prompt, and therefore cannot eliminate all model-specific regressions.

The DeepSeek-V4 routing baseline further improves the always-rewrite variant, reaching or exceeding the original-prompt score on several generators. Its decisions nevertheless remain weaker than those of the learned aligner on all seven evaluated backbones. In contrast, the full system raises the alignment score over the original prompt from 0.6544 to 0.6613 on SD2.1, 0.6591 to 0.6669 on CoMat 2.1, 0.7379 to 0.7412 on SD3, 0.7579 to 0.7626 on Flux.1, 0.7536 to 0.7594 on PixArt- Σ , 0.7182 to 0.7335 on Infinity, and 0.6847 to 0.7014 on Show-O2. The larger gains on Infinity and Show-O2, together with the smaller but consistent gains on SD3 and Flux.1, indicate that prompt preference spaces differ across generators. The target-aware aligner is thus most useful as a model-specific safeguard against harmful rewriting, rather than merely as another general-purpose prompt optimizer.

Human Evaluation

To complement quantitative evaluations with a human-centered assessment of intent preservation, we conducted a user study comparing our method with existing approaches. We first sampled user intents and applied different optimization methods to obtain model-specific refined prompts, which were then used to generate images. A total of 20 trained experts with domain-specific professional backgrounds participated. For image comparisons, annotators were shown the stated user intent together with the candidate images and answered: “Which image better preserves the stated user intent?” For prompt comparisons, they selected the candidate prompt that more faithfully preserved the same stated intent. The task therefore measures intent faithfulness rather than visual appeal or unconstrained preference. As shown in Figure 3, NoxEye was selected most frequently, supporting its ability to preserve user intent through rewriting and generation.

Inference Efficiency

The deployed aligner operates entirely on the original prompt and its candidate rewrite. It computes surface statistics, Qwen text embeddings, and the target generator’s text-encoder embeddings, and then applies a lightweight classical regressor. These are all prompt-side operations. In particular, the eight paired generations and Qwen3-VL image scoring used to construct the training targets are *not* repeated at infer-

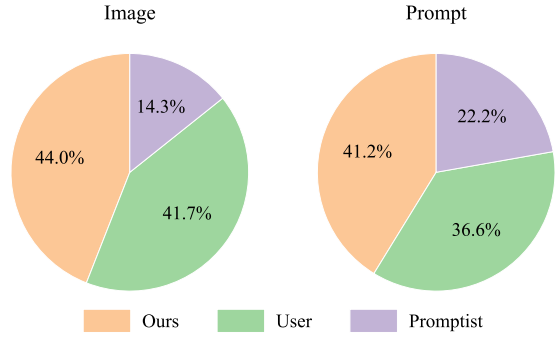


Figure 3: Human evaluation of intent faithfulness. Annotators selected the image or prompt that better preserved the stated user intent; NoxEye was compared with the user prompt and Promptist (Hao et al. 2023).

ence time. Routing therefore introduces no additional image-generation latency and requires neither generated images nor an image-side MLLM call. The remaining overhead consists only of producing the candidate rewrite and evaluating its text features; once the route is chosen, the target T2I model is invoked exactly once with either P or P^* .

Additional Qualitative Results

Examples for the Three Failure Modes

We present more visual results between the images generated with different prompts. As illustrated in the Figure 4, when a prompt simultaneously mentions both *fish* and *cat*, the generated images may omit the fish, indicating **informational sparsity**, where overly concise user prompts encourage LLM-based expansion to hallucinate unintended attributes and deviate from user intent. In addition, anomalous fish–cat amalgamations are observed, revealing **lexical perturbation and noise sensitivity**, in which minor word-level disturbances can mislead the text encoder and result in inconsistent or even contradictory generations.

Rewrite-versus-Retain Examples

The following cases, selected from the AP Bench analysis artifacts, illustrate why routing must be performed per prompt and per target generator. Scores are Qwen3-VL semantic-alignment scores averaged over the eight matched generations used by the evaluation pipeline.

Rewrite: recovering explicit object composition. For the novice prompt “Groceries, like assorted milk bottle, bread packed, soft drink, and some assorted vegetables in recyclable bag to carry,” the enhancer produces “A shopping bag containing milk, bread, soft drinks, and assorted vegetables.” The rewrite removes grammatical noise while retaining the complete object list and container relation. Its score increases from 0.5813 to 0.7625 on SD2.1 and from 0.6313 to 0.7563 on Flux.1. This is a representative rewrite case: canonicalization makes the intended composition easier for both generators to recover without adding a new semantic attribute.

Retain: preserving an already informative prompt. The prompt “Full image of a massive white [sic] palace with an aquamarine roof, built on a beach, Italian architecture, highly detailed, Unreal Engine, 8D” already specifies the subject, roof color, location, architectural style, and rendering cues. Its concise rewrite—“Massive white palace with aquamarine roof, Italian architecture, highly detailed, built on a beach, Unreal Engine 8D”—does not add useful information. On SD2.1, the original scores 0.8688 while the rewrite scores 0.8438; on Flux.1 both score 0.9500. Retaining the original avoids a small regression while sacrificing no measured gain on the stronger generator.

Generator-dependent routing. For “a soda bottle covered in water drops, on ice, ice background,” the rewrite “A high-quality masterpiece image of a soda bottle on ice with water drops” improves SD2.1 from 0.3000 to 0.4063, but slightly decreases Flux.1 from 0.3563 to 0.3438. The appropriate decision is therefore not a global property of the prompt pair: it also depends on how the target generator encodes the retained and reorganized information. This observation motivates the target-text-encoder features used by the aligner.

Failure Cases and Limitations

The selector is not an oracle. The validation-selected regressors exhibit only modest rank correlation with realized per-prompt gains, and the aggregate oracle gap shows that some beneficial rewrites are retained while some harmful rewrites are selected. Three recurring limitations are visible in the analysis artifacts.

First, *generator-specific reversals* remain difficult. In the soda-bottle case above, the same rewrite helps SD2.1 but is slightly harmful to Flux.1. Text-encoder features reduce this ambiguity but cannot fully predict the stochastic image-side response. Second, *specialized prompt syntax* can be semantically important even when it appears noisy. For “Pickup Truck, mushroomz, <lora:mushroomz:1>,” the intended concept is a vintage pickup truck merged with a giant mushroom. Replacing the LoRA-style syntax with fluent prose can instead describe a truck merely surrounded by mushrooms, changing the central compositional relation. Such adapter tokens and model-specific control syntax are not reliably represented by generic text embeddings. Third, *metric uncertainty* remains: Qwen3-VL is an operational alignment proxy and may overlook subtle style, spatial, or identity-preservation errors. Averaging eight matched generations reduces sampling noise in training labels but does not remove systematic evaluator bias.

More broadly, routing is limited to the original prompt and one enhancer candidate; it cannot recover a better formulation outside that candidate set. The selected regressor and threshold are validation-dependent, so performance may shift for new generators, languages, prompt conventions, or user domains. These cases motivate future work on syntax-aware features, calibrated uncertainty or abstention, and routing among multiple intent-preserving candidates rather than a single binary choice.

Ethical Considerations and Dataset Limitations

AP Bench is constructed from publicly available prompt-image pairs shared on online text-to-image communities, including Lexica and Civitai. We only collect information required for benchmark construction, such as user prompts, associated images, and available generation metadata, while excluding personally identifiable information. The collected content is used solely for academic research on prompt informativeness and generation stability. During data verification, we manually filter inappropriate or sensitive content; however, automatic and human-based filtering cannot guarantee complete removal of all potentially problematic samples.

Several limitations should be considered when interpreting our benchmark and evaluation results. First, user intent is inherently complex and cannot be fully represented by a single category or prompt-image pair. Although AP Bench leverages observable intent signals from authentic user prompts and images, it primarily measures intent preservation rather than recovering the complete latent intention of users. In addition, the initial taxonomy is inspired by ImageNet categories, which introduces an object-centric bias. While we include out-of-taxonomy samples to alleviate this limitation, future work may extend the benchmark toward broader semantic domains, abstract concepts, and multilingual prompting scenarios.

Second, AP Bench is collected from existing text-to-image sharing platforms, whose users may not represent the diversity of general image-generation users. The benchmark may therefore inherit biases from platform-specific user behaviors, artistic preferences, and content distributions. Moreover, our evaluation relies on CLIP, MLLM-based semantic scoring, and language-model-based entropy and energy measurements. These metrics provide complementary operational signals but may contain biases from their pretraining data and should not be interpreted as absolute measures of prompt quality or user preference.

Finally, prompt optimization systems may unintentionally modify user intent or introduce unsupported information. This motivates our selective refinement design, which aims to determine whether rewriting is beneficial rather than assuming universal improvement. Nevertheless, NoxEye remains a learned model and may still produce incorrect refinements in some cases; human oversight remains important for applications where preserving individual creativity and expression is critical.

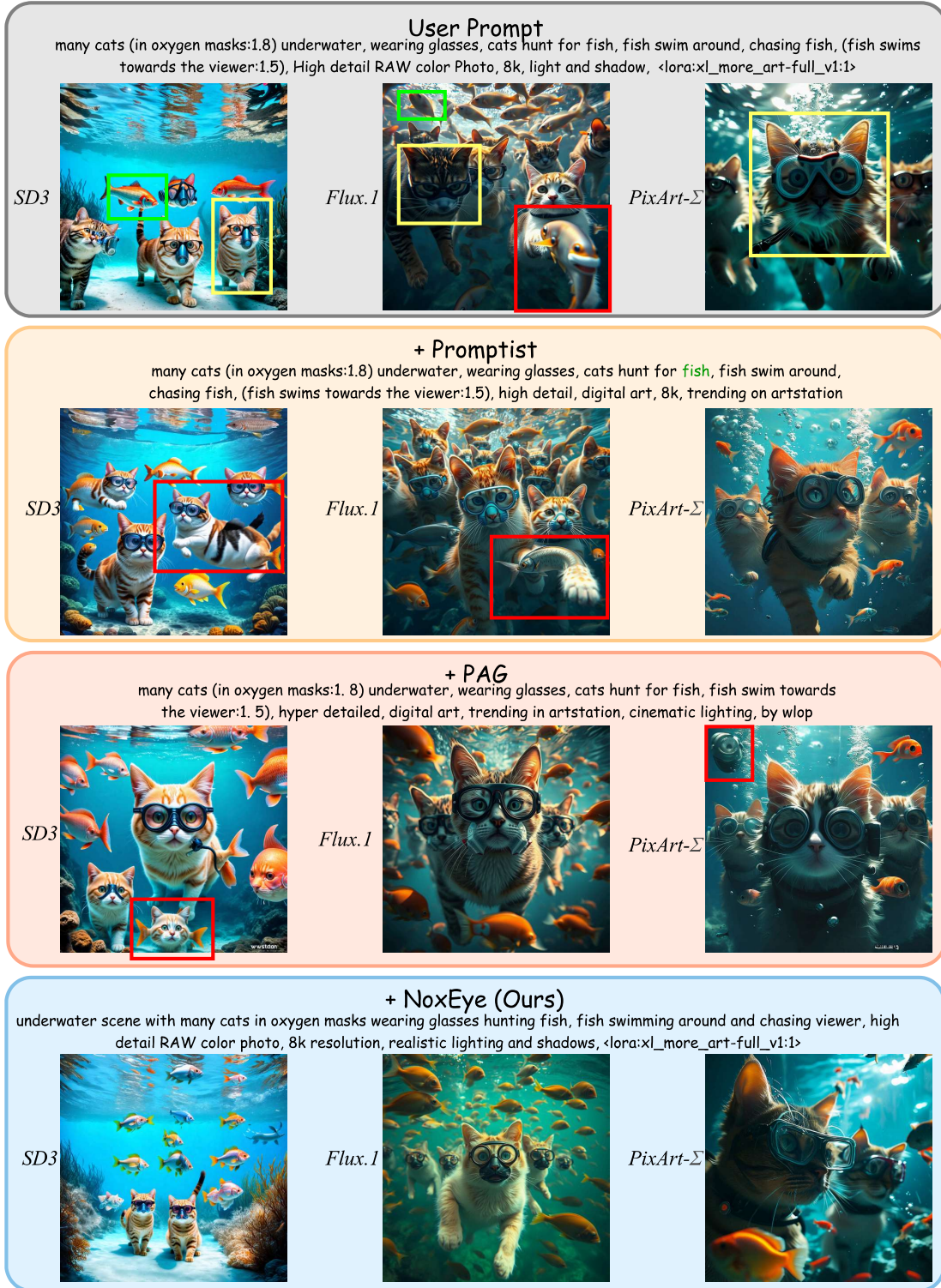


Figure 4: **Informational sparsity** stems from user prompts, while **lexical perturbation** and **noise sensitivity** arise in other prompt optimization approaches.

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